A Cantilever loaded at its free end:

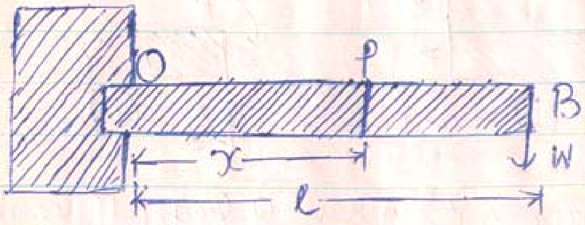


fig. 1

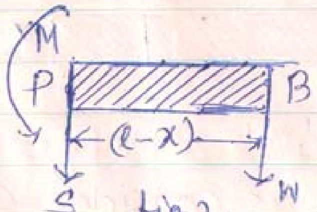


fig 2.

A Cantilever is a light beam (of negligible weight) of uniform cross section throughout, clamped rigidly at one end.

The clamped end O' has been taken as origin.

The undisturbed position of the Cantilever is taken as X-axis.

Given: W = The Load Suspended at its free end B.

l = The length OB of the Cantilever.

Y = Youngs modulus of the material of the Cantilever.

A = Area of cross-section of the Cantilever.

Let us consider any point P at a distance x from the origin.

Let us imagine a section of the beam at P.

The portion PB is shown separately in fig 2.

When the Cantilever is bent a system of forces acts on it, which is equivalent to a force and a couple. According to conversion, the left hand portion OP exerts on the right hand portion PB; a shearing force S along downward direction and a bending moment M along anticlockwise direction.

Since the portion PB is in equilibrium.

$$\sum F = 0 \quad \& \quad \sum T = 0$$

i.e $\sum F = \cancel{S + W + n}$ $\quad S + W = 0$ or $\boxed{S = -W} \longrightarrow (1)$

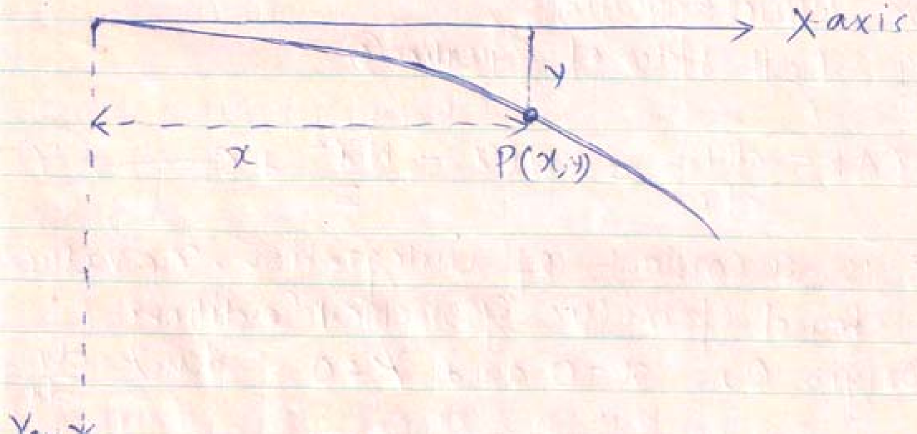
The negative sign indicates that the shearing force at a point P must be along upward direction.

$\sum T$ = Taking moment about P

$$= -M + W(l - x) = 0$$

−ve sign indicates anticlockwise moment

$$\text{or } M = W(l - x) \longrightarrow (2)$$



Let R = The radius of curvature of bending, at the point P.

∴ Bending moment at P : $M = \dfrac{YAK^2}{R} \longrightarrow (3)$

Equating (3) and (2) :

$$\frac{YAK^2}{R} = W(l - x) \longrightarrow (4)$$

From Calculus we know that the curvature P

$$\rho = \frac{1}{R} = \frac{d^2y/dx^2}{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}$$

But for small curvature :

$\tan\psi = \frac{dy}{dx}$ is very small ; hence its

Square can be neglected to one.

For small curvature ; $\frac{1}{R} = \frac{d^2y}{dx^2} \longrightarrow (5)$

Putting equation (5) in (4)

$$YAk^2 . \frac{d^2y}{dx^2} = W(l-x) \longrightarrow (6)$$

Equation (6) is a second order differential equation we now solve this equation :-

Integrating both sides of equation (6) :

$$YAk^2 \frac{dy}{dx} = Wlx - \frac{Wx^2}{2} + C \longrightarrow (7)$$

Where C is a constant of integration. The value of C can be found from the geometrical condition.

At the origin O ; $x = 0$ and $\psi = 0$ $\therefore \tan\psi = \frac{dy}{dx} = 0$

hence from (7) $\boxed{C = 0} \longrightarrow (8)$ Putting equation (8) in (7) :

$$YAk^2 \frac{dy}{dx} = Wlx - \frac{Wx^2}{2} \longrightarrow (9)$$

Integrating both sides of equation (9) :-

$$YAk^2 y = \frac{Wlx^2}{2} - \frac{Wx^3}{2 \cdot 3} + D \longrightarrow (10)$$

Where D is an unknown constant of integration.
Applying the geometrical condition at the origin O
$x=0,\ y=0$ from (10), $D=0 \longrightarrow$ (11)
Putting (11) in (10) :-

$$YAk^2y = \frac{Wlx^2}{2} - \frac{Wx^3}{6} \longrightarrow (12)$$

Using equation (12), we can find the depression y at any point, at a distance x from the origin.

Let δ be the depression at the loaded end B
at B, $x=l$, $y=\delta$

from (12) :- $YAk^2\delta = \frac{Wl\,l^2}{2} - \frac{Wl^3}{6} = \frac{Wl^3}{2} - \frac{Wl^3}{6}$

$$Y = \frac{Wl^3}{3AK^2\delta} \longrightarrow (13)$$

If the beam is of rectangular cross-section and 'b' & 'd' be the length of the horizontal and vertical sides of the cross section

$$AK^2 = \frac{bd^3}{12} \longrightarrow (14)$$

Putting (14) in (13) :-

$$Y = \frac{Wl^3}{3 \cdot \frac{bd^3}{12} \cdot \delta} = \frac{4Wl^3}{bd^3\delta} \longrightarrow (15)$$

Young's modulus of the material of the cantilever, can be calculated by measuring the depression δ at its loaded end.

Show that for a light Cantilever, the depression at any point P, when loaded at any other point Q is same as the depression at Q, when loaded at P.

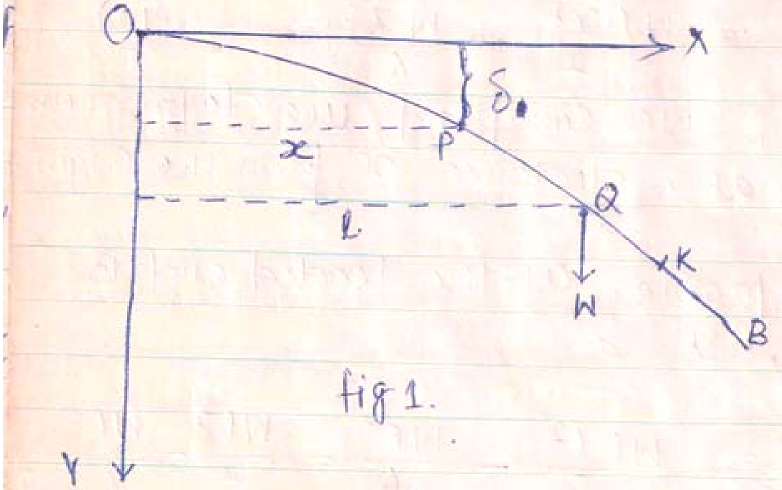


fig 1.

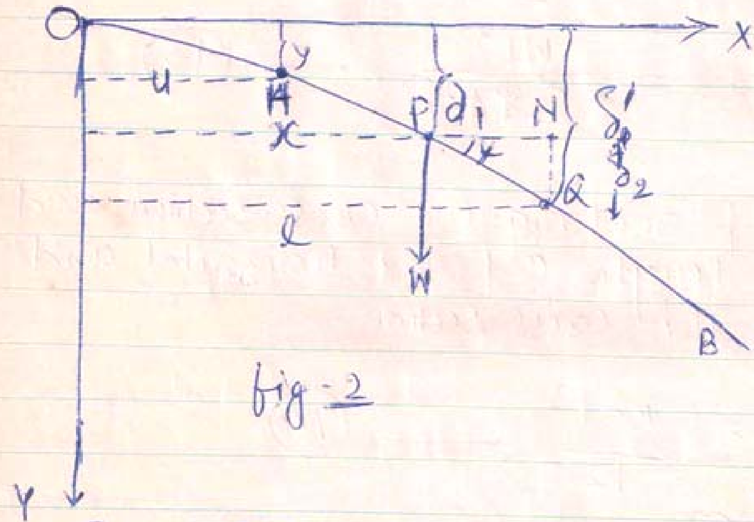


fig - 2

Given : OB is a light Cantilever. OB = L = length of the Cantilever.

$\boxed{OQ = l}$ $\boxed{OP = x}$ where P and Q are the two chosen points

W = load which is suspended

The undisturbed position of the Cantilever is taken as X-axis and a perpendicular to X-axis along downward direction is taken as Y-axis

Case I : The Cantilever is loaded at the point Q, to find the depression at P.

Let us consider any point 'K' between Q and B and let us imagine a section at K, the portion KB is shown separately in fig 3.

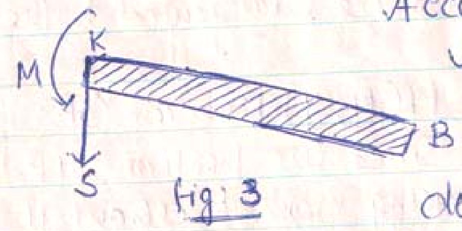


fig 3

According to our convension, the left hand portion OK, exerts on the right hand portion KB a shearing force 'S' along downward direction and a bending moment M along anticlockwise direction as shown.

Since the portion KB is in equilibrium

$$\sum F = 0 \ \& \sum T = 0 \quad \text{ie} \quad \sum F = S = 0$$

$$\sum T = M = 0 \quad \therefore \boxed{M = 0} \quad \text{or} \quad \frac{YAK^2}{R} = 0$$

Where R = Radius of curvature of bending at the point K.

$$\frac{YAK^2}{R} = 0 \Rightarrow \frac{1}{R} = 0 \quad \boxed{\text{or } R = \infty}$$ Thus the radius of curvature of any point between Q & B is infinity i.e the portion QB is a straight line, tangential to the curve OQ at Q.

Hence the problem reduces to the problem of a Cantilever loaded at its free end Q.

Hence using the result of the problem, the depression δ at any point 'P' at a distance x

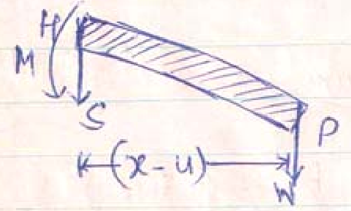
$$YAK^2 \delta = Wl\frac{x^2}{2} - \frac{Wx^3}{6} \quad \text{-----} \ (\text{a})$$

$$\delta = \frac{Wx^2}{YAK^2}\left(\frac{l}{2} - \frac{x}{6}\right) \longrightarrow (1)$$

Case II The load W is suspended at P and we have to find the depression at Q (fig-2).

Considering any point between P & B and following a similar reasoning as above we can say that, the portion PB is a straight line, tangential to the curve OP at P.

Consider any point H(u, y) between OP. Considering a section at H, since the portion HP is in equilibrium $\Sigma T = 0$. Taking moment about H



$$M - W(x-u) = 0$$

$$\text{or } M = W(x-u)$$

$$\text{or } \frac{YAK^2}{R} = W(x-u)$$

$$\frac{1}{R} = \frac{d^2y}{du^2}$$

$$\text{or} \quad YAK^2 \frac{d^2y}{du^2} = W(x-u)$$

Integrating both sides :-

$$YAK^2 \frac{dy}{du} = Wxu - \frac{Wu^2}{2} + c' \longrightarrow (2)$$

Where c' is a constant of integration. Applying Geometrical condition at the origin O.

$\boxed{u = 0}$ & $\boxed{\frac{dy}{du} = 0}$ from (2) $c' = 0$

Putting the value of c' in (2) :

$$YAK^2 \frac{dy}{du} = Wxu - \frac{Wu^2}{2} \longrightarrow (3)$$

let the tangent PB at the point P of the curve OP make an angle χ with x-axis

$$\therefore \tan\chi = \left[\frac{dy}{du}\right]_P = \left[\frac{dy}{du}\right]_{u=x} = \frac{1}{YAk^2}\left[Wx^2 - \frac{Wx^2}{2}\right]$$ from (3):

$$\therefore \tan\chi = \left[\frac{dy}{du}\right]_{u=x} = \frac{1}{YAk^2}\frac{Wx^2}{2} \longrightarrow (4)$$

Integrating both sides of Equation (3)

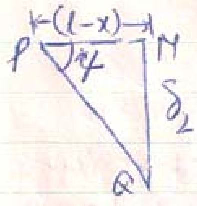
$$YAk^2 y = Wx\frac{u^2}{2} - \frac{Wu^3}{6} + D' \longrightarrow (5)$$

at the origin O; $u=0$ & $y=0$ from (5) $D'=0$

Equation (5) reduces to $YAk^2 y = Wx\frac{u^2}{2} - \frac{Wu^3}{6} \longrightarrow (6)$

let δ_1 be the depression at P
ie $\boxed{u=x}$ $\boxed{y=\delta_1}$ from (6): $YAk^2\delta_1 = \frac{Wx^3}{2} - \frac{Wx^3}{6}$

$$\text{or } \delta_1 = \frac{1}{YAk^2}\frac{Wx^3}{3} \longrightarrow (7)$$



from $\triangle$ PQM: $\tan\chi = \frac{\delta_2}{(l-x)}$

$$\delta_2 = (l-x)\tan\chi$$

Putting Equation (4):

$$\delta_2 = \frac{(l-x)\,Wx^2/2}{YAk^2} \longrightarrow (8)$$

Adding (7) and (8): we get the total depression δ'

at the point Q, when loaded at P:

$$\therefore \delta' = \delta_1 + \delta_2 = \frac{1}{YAK^2}\frac{Wx^3}{3} + \frac{(l-x)Wx^2/2}{YAK^2}$$

$$\delta' = \frac{Wx^2}{YAK^2}\left[\frac{x}{3} + \frac{l}{2} - \frac{x}{2}\right]$$

$$= \frac{Wx^2}{YAK^2}\left[\frac{l}{2} - \frac{x}{6}\right]$$

$$\text{or } \delta' = \frac{Wx^2}{YAK^2}\left[\frac{l}{2} - \frac{x}{6}\right] \longrightarrow (9)$$

$$\text{from (1)}: \delta = \frac{Wx^2}{YAK^2}\left[\frac{l}{2} - \frac{x}{6}\right] \longrightarrow (10)$$

Comparing (9) and (10) : $\boxed{\delta = \delta'}$ Proved

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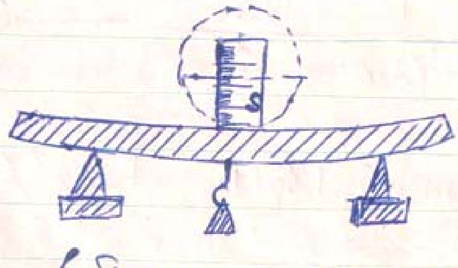

(Determination of Y by non uniform bending of beam; the deflection is either measured accurately by microscope or directly by spherometer).